

## **THEORIES OF FAILURE**

### **Introduction**

#### **Necessity for a Theory:**

Many hypotheses have been presented as to the conditions which determine elastic failures of metals subjected to steady constant stresses. Due to the complex nature of failure no uniformity of opinion has been reached so far. From the physical point of view, failure of a material represents either direct separation of particles from each other (Brittle failure) or slipping of particles (Ductile failure) accompanied by plastic deformation. A material is said to have failed if the stress induced has crossed the elastic limit and permanent deformation has taken place.

According to the chief theories, perfect elasticity ceases when a certain limiting value is reached by one of the following theory:

- (1) The maximum principal stress theory
- (2) The maximum principal strain theory
- (3) The maximum shear stress theory
- (4) The maximum strain energy theory
- (5) The maximum shear strain energy theory.

### **7.2 Maximum Principal Stress Theory ( Rankine's Theory )**

It is the simplest and the oldest theory of failure. According to this theory permanent failure takes place under a state of stress when the value of the maximum principal stress reaches the value of the yield point stress as found in a simple tensile test or minor principal stress (the compressive stress) is equal to yield point stress in simple compression.

Hence the condition of yielding is

$$\sigma_t = \sigma_y \quad \text{.....(7.1)}$$

$$|\sigma_c| = \sigma'_y \quad \text{.....(7.2)}$$

Where  $\sigma_y$  - yield point stress in simple tension

$\sigma'_y$  - yield point stress in simple compression.

If maximum principal stress is the design criterion the maximum principal stress must not exceed the working stress  $\sigma$  for the material. Hence

$$\sigma_1 \leq \sigma \quad \dots (7.3)$$

**Comments:**

- i) This theory does not take the effect of other principal stresses and of the shearing stresses on other planes through the element.
- ii) For brittle materials which do not fail by yielding but fail by brittle fracture, the maximum principal stress theory is considered satisfactory.

**7.3 Maximum Principal Strain Theory (Saint Venant):**

Ductile material begin to yield when the principal strain reaches the strain at which yielding occurs in simple tension or when the maximum principal strain (compression strain) equals the yield point strain in simple compression.

In a 3D stress system (Fig. 12.1) the strains are as follows

$$e_1 = \frac{\sigma_1}{E} - \frac{1}{mE}(\sigma_2 + \sigma_3)$$

$$e_2 = \frac{\sigma_2}{E} - \frac{1}{mE}(\sigma_3 + \sigma_1)$$

$$e_3 = \frac{\sigma_3}{E} - \frac{1}{mE}(\sigma_1 + \sigma_2)$$

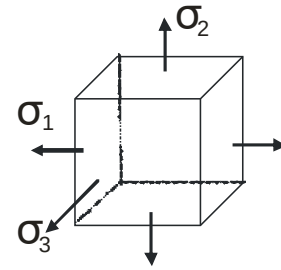


Fig. 7.1

If  $e_y = \text{yield point strain (tensile)} = \frac{\sigma_y}{E}$

$e'_y = \text{yield point strain (compressive)} = \frac{\sigma'_y}{E}$

then  $e_1 = e_y$

or  $\left| \sigma_1 - \frac{1}{m}(\sigma_2 + \sigma_3) \right| = \sigma_y \quad \dots (7.4a)$

and  $e_3 = e'_y$

or  $\left| \sigma_3 - \frac{1}{m}(\sigma_1 + \sigma_2) \right| = \sigma'_y \quad \dots (7.4b)$

In a 2D stress system having  $\sigma_1$  and  $\sigma_2$  as the major and minor principal stresses equation (12.4a) reduces to

$$\sigma_1 - \frac{\sigma_2}{m} = \sigma_y \quad \dots (7.4 c)$$

Thus the stress which acting alone produces the same maximum strain  $e_1$  equals to  $\sigma_1 - \frac{\sigma_2}{m}$ .

The design criterion as per this theory is

$$\sigma_1 - \frac{\sigma_2}{m} \leq \sigma \quad \dots (7.5)$$

where  $\sigma$  is the working stress of the material.

**Comments:**

- i) This theory considers all principal stresses hence it is considerable for brittle material.
- ii) This theory is invalid when

$$\sigma_1 = \sigma_2$$

i.e. equal tension in two perpendiculars directions

we have 
$$\sigma_1 - \frac{\sigma_1}{m} = \sigma_y$$

or 
$$\left( \frac{m-1}{m} \right) \sigma_1 = \sigma_y$$

this gives 
$$\sigma_1 > \sigma_y$$

Thus this theory gives that the tensile stress at yielding will be higher than the stress in simple tension which is not supported by the experiments.

**7.4 Maximum Shear Stress Theory (Guest's Theory)**

As per this yielding begin when the maximum shearing stress equals the maximum shearing stress developed at the yielding in simple tension. This theory results from the observation that in a ductile material, slip occurs during yielding along critically oriented planes. This suggests that the maximum shear stress plays the key role. Since the maximum shear stress is equal to half the difference between the maximum and minimum principal stresses and the maximum shear in simple tension is equal to half the tensile stress, we have

$$\tau_{\max} = \frac{1}{2}(\sigma_1 - \sigma_3) = \frac{\sigma_y}{2}$$

Hence the condition of yielding is

$$\sigma_1 - \sigma_3 = \sigma_y \quad \dots (7.6)$$

In case of 2D stresses system  $\sigma_3 = 0$ ,  $\sigma$  is working stress the design criterion is

$$(\sigma_1 - \sigma_2) \leq \sigma \quad (\text{Provided } \sigma_2 < 0 \text{ comp.}) \quad \dots (7.7)$$

**Comments:**

- i) This theory is good for ductile materials but not good enough for the state of stress consisting tri axial tensile stresses of nearly equal magnitude, reducing the shearing stress to small magnitude so that failure would be by brittle fracture rather than by yielding.
- ii) This theory does not give satisfying results for the state of stress of pure shear in which maximum amount of shearing is developed. (i.e. torsion test).

**7.5 Maximum Strain Energy Theory (Haigh's Theory):**

This theory that inelastic action at any point in a body due to any state of stress begins only when the energy per unit volume absorbed at the point is equal to the energy absorbed per unit volume of the material when subjected to the elastic limit in simple tensile test. In a 3D stress system the strain energy per unit volume is given by

$$U = \frac{1}{2E} \left[ \sigma_1^2 + \sigma_2^2 + \sigma_3^2 - \frac{2\sigma_1\sigma_2 + 2\sigma_2\sigma_3 + 2\sigma_3\sigma_1}{m} \right]$$

(where  $\sigma_1, \sigma_2, \sigma_3$  the same sign)

Hence yield criterion can be represented as

$$U \leq \frac{\sigma_y^2}{2E}$$

$$\text{or} \quad \sigma_1^2 + \sigma_2^2 + \sigma_3^2 - \frac{2\sigma_1\sigma_2 + 2\sigma_2\sigma_3 + 2\sigma_3\sigma_1}{m} = \sigma_y^2 \quad \dots (7.8)$$

Taking a 2D case ( $\sigma_3 = 0$ ) equation 12.8 reduces to

$$\sigma_1^2 + \sigma_2^2 - \frac{2\sigma_1\sigma_2}{m} = \sigma_y^2$$

If  $\sigma$  is the working stress the design criterion may be stated as

$$\sigma_1^2 + \sigma_2^2 - \frac{2\sigma_1\sigma_2}{m} \leq \sigma^2 \quad \dots (7.9)$$

**Comments:**

This theory is not applicable to brittle materials as those have different elastic limit in tension and compression.

## 7.6 Max. Shear Strain Ensergy Theory (Miss Henkly Theory)

### **Max distortion energy theory**

This theory states that inelastic action at any point in a body under any stress system begins only when the strain energy of distortion per unit absorbed at the point is equal to the strain energy of distortion per unit volume at any point in a bar stressed to the elastic limit under a state of simple tension (or compression) test.

In above approach the total elastic energy is divided into two parts, one associated with the volumetric changes of the material and other causing shear distortions. By equating the shear distortion energy at yield point in simple tension to that under combined stress, the yield criterion for combined stress is established.

Under a system of stresses acting externally the total work done causes:

- (a) Change of volume due to operation of direct stress
- (b) Distortion due to shearing stress which does not effect the volumetric change.

Thus total  $U = U_v + U_s$

$$\text{Now } U_v = \frac{1}{2}(\text{average } \sigma) \Delta V = \frac{\sigma_1 + \sigma_2 + \sigma_3}{6} \Delta V$$

$$\begin{aligned} \text{Change in volume } \Delta V &= \frac{1}{E} \left\{ \sigma_1 + \sigma_2 + \sigma_3 - \frac{2}{m} (\sigma_1 + \sigma_2 + \sigma_3) \right\} \\ &= \frac{1}{E} (\sigma_1 + \sigma_2 + \sigma_3) \left( \frac{m-2}{m} \right) \end{aligned}$$

$$\begin{aligned} U_v &= \frac{m-2}{6mE} \{ \sigma_1 + \sigma_2 + \sigma_3 \}^2 \\ &= \frac{m-2}{6mE} \left[ \sigma_1^2 + \sigma_2^2 + \sigma_3^2 + 2(\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1) \right] \\ &= \frac{m-2}{6mE} [X + 2Y] \end{aligned} \quad \dots\dots(7.10)$$

( Where  $X = \sigma_1^2 + \sigma_2^2 + \sigma_3^2$ ,  $y = \sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1$  )

$$\begin{aligned} \text{And } U &= \frac{1}{2E} \left[ \sigma_1^2 + \sigma_2^2 + \sigma_3^2 - \frac{2}{m} (\sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1) \right] \\ &= \frac{1}{2} E \left( X - \frac{2}{m} Y \right) \end{aligned} \quad \dots\dots(7.11)$$

Hence  $U_s = U - U_v$

$$\begin{aligned}
&= \frac{1}{2E} \left( X - \frac{2}{m} Y \right) - \frac{m-2}{6mE} (X + 2Y) \\
&= \frac{1}{6E} \left( 3X - \frac{6Y}{m} - X - 2Y + \frac{2X}{m} + \frac{4Y}{m} \right) \\
&= \frac{1}{3E} \left[ X \left( 1 + \frac{1}{m} \right) - Y \left( 1 + \frac{1}{m} \right) \right] \\
U_s &= \frac{m+1}{3mE} (X - Y) \\
&= \frac{m+2}{6mE} (\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - \sigma_1\sigma_2 + \sigma_2\sigma_3 + \sigma_3\sigma_1) \\
&= \frac{m+2}{6mE} ((\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2) \quad \dots\dots(7.12)
\end{aligned}$$

The above expression is for strain energy for distortion.

Under the action of an axial stress  $\sigma_y$  the strain energy of distortion will be

$$\frac{m+2}{6mE} \sigma_y^2$$

Hence the failure criterion may be stated as follows:

$$\begin{aligned}
&\frac{m+1}{6mE} \{ (\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2 \} = \frac{m+2}{3mE} \sigma_y^2 \\
\text{or} \quad &\frac{1}{2} (\sigma_1 - \sigma_2)^2 + \frac{1}{2} (\sigma_2 - \sigma_3)^2 + \frac{1}{2} (\sigma_3 - \sigma_1)^2 = \sigma_y^2 \quad \dots\dots(7.13)
\end{aligned}$$

in a 2D system ( $\sigma_3 = 0$ )

$$\sigma_1^2 + \sigma_2^2 - \sigma_1\sigma_2 = \sigma_y^2 \quad \dots\dots (7.14)$$

If  $\sigma$  is the working stress in the material the design criterion will be

$$\sigma_1^2 + \sigma_2^2 - \sigma_1\sigma_2 \leq \sigma^2 \quad \dots\dots (7.15)$$

### Comments:

This theory is not applicable to brittle materials as those have different elastic limit in tension and compression.